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1: Simultaneous Equations

2: Simultaneous Equations using Graphs

3: Solving Inequalities

4: Sketching Inequalities

The solution(s) to an equation may be:

A single value:	$2x + 1 = 5$
Multiple values:	$x^2 + 3x + 2 = 0$
An infinitely large set of values:	$x > 3$
No (real) values!	$x^2 = -1$
Every value!	$x^2 + x = x(x + 1)$

The point is that you shouldn't think of the solution to an equation/ inequality as an 'answer', but a **set** of values, which might just be a set of 1 value (known as a singleton set), a set of no values (i.e. the empty set $\varnothing$), or an infinite set (in the last example above, this was $\mathbb{R}$)

✎ The solutions to an equation are known as the **solution set**.

For simultaneous equations, the same is true, except each 'solution' in the solution set is an assignment to **multiple** variables.
All equations have to be satisfied **at the same time, i.e. 'simultaneously'**.

Scenario	**Example**	**Solution Set**
A single solution:		
Two solutions:		
No solutions:		
Infinitely large set of solutions:		

Recap!

Solve the simultaneous equations:

$3x + y = 8$

$2x - 3y = 9$

We can either use substitution (i.e. making x or y the subject of one equation, and substituting it into the other) or elimination, but the latter is easier for linear equations.

9x+3y=24

2x−3y=9

:

Solve the simultaneous equations:

$x + 2y = 3$

$x^2 + 3xy = 10$

We can't use elimination this time as nothing would cancel.

We instead:

(1) Rearrange linear equation to make x or y the subject.

(2) Substitute into quadratic equation and solve.

Solve the simultaneous equations:

$3x^2 + y^2 = 21$

$y = x + 1$

4 Solve the simultaneous equations:
$x + y = 3, \quad x^2 - 3y = 1$

5 a By eliminating y from the equations
$y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, show that $x^2 - 2x - 11 = 0$.
b Hence, or otherwise, solve the simultaneous equations
$y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, giving your answers in the form
$a \pm b\sqrt{3}$, where a,b are integers.

6 One pair of solutions for the simultaneous equations $y = kx - 5, \quad 4x^2 - xy = 6$ is (1,p) where k,p are co

a\ Find the values of k,p.

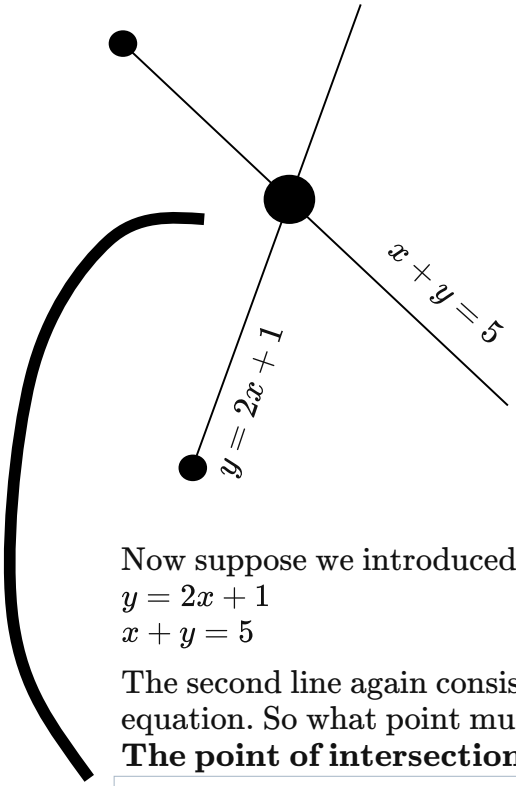
b Find the second pair of solutions for the simultaneous equations.

Extension

1 *[MAT 2012 1G]* There are *positive* real numbers x and y which solve the equations
$2x + ky = 4$,
$x + y = k$
for:
A) All values of k;
B) No values of k;
C) $k = 2$ only;
D) Only $k > -2$

If k=2 then 2x+2y=4 and x+y=2 which are equivalent.

2 *[STEP 2010 Q1]* Given that
$$5x^2 + 2y^2 - 6xy + 4x - 4y \equiv a(x - y + 2)^2 + b(cx + y)^2 + d$$
a) Find the values of a, b, c, d.
b) Solve the simultaneous equations:
$5x^2 + 2y^2 - 6xy + 4x - 4y = 9$,
$6x^2 + 3y^2 - 8xy + 8x - 8y = 14$
(Hint: Can we use the same method in (a) to rewrite the second equation?)



Recall that a line with a given equation is **the set of all points that satisfy the equation**.

i.e. It is a graphical representation of the solution set where each (x, y) point represents each of the solutions x and y to the equation.
e.g. $x = 3, y = 7$

Now suppose we introduced a second simultaneous equation:
$y = 2x + 1$
$x + y = 5$

The second line again consists of all points (x, y) which satisfy the equation. So what point must satisfy both equations simultaneously?
The point of intersection!

Example 12

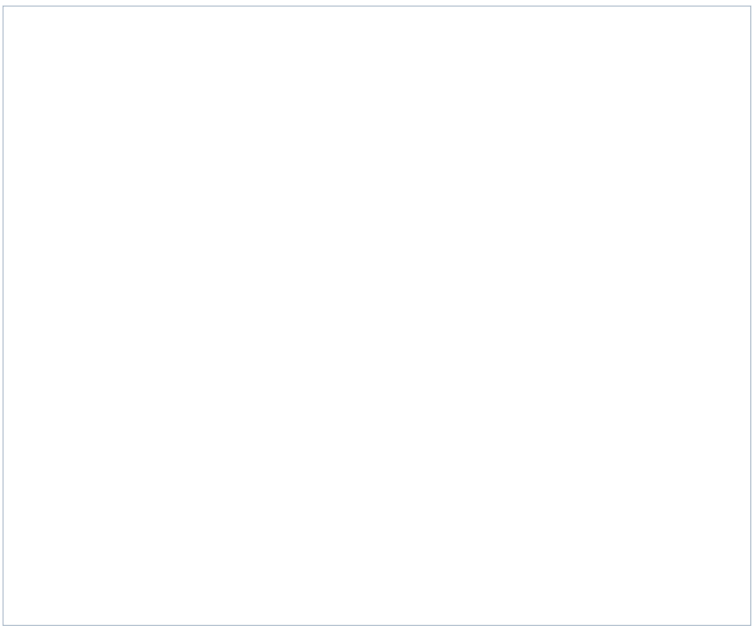
a) On the same axes, draw the graphs of:
$2x + y = 3$
$y = x^2 - 3x + 1$

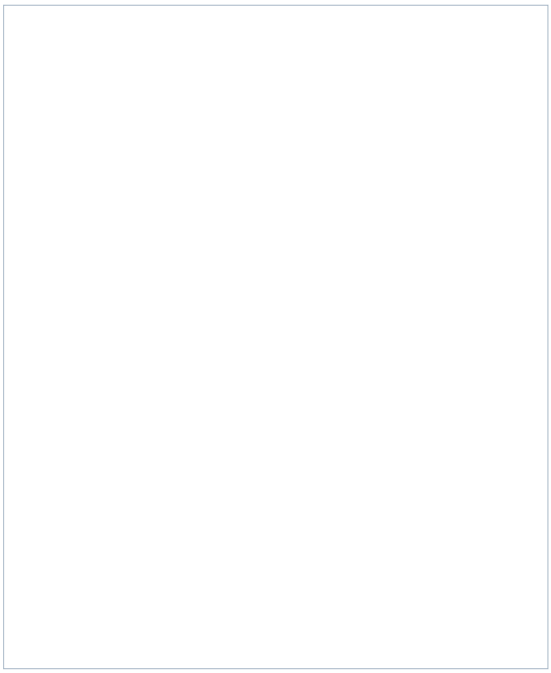
b) Use your graph to write down the solutions to the simultaneous equations.

c) What algebraic method (perhaps thinking about the previous chapter), could we have used to show the graphs would have intersected twice?

Substituting linear equation into quadratic:

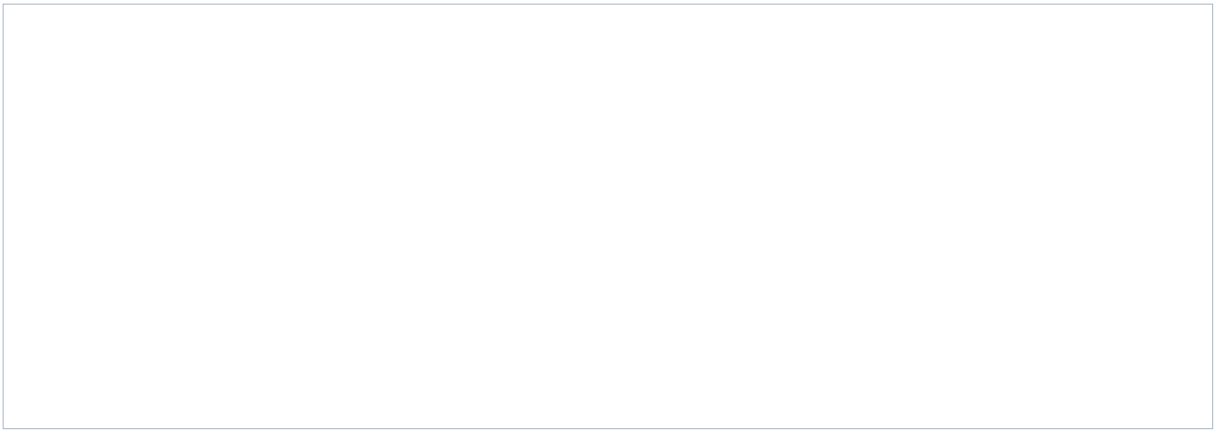
a) On the same axes, draw the graphs of:
$y = 2x - 2$
$y = x^2 + 4x + 1$



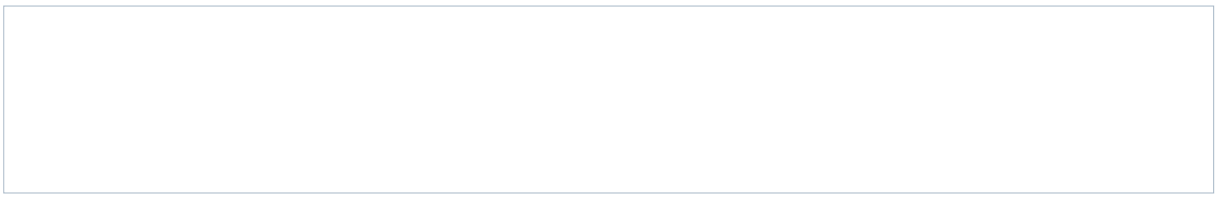


The line with equation $y = 2x + 1$ meets the curve with equation $kx^2 + 2y + (k - 2) = 0$ at exactly one point. Given that k is a positive constant:
a) Find the value of k.
b) For this value of k, find the coordinates of this point of intersection.

a



b



8 Given the simultaneous equations $2x - y = 1, \quad x^2 + 4ky + 5k = 0,$
where k is a non-zero constant,

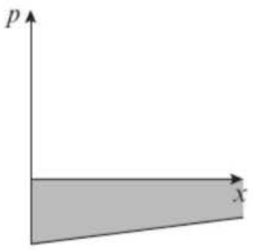
a show that $x^2 + 8kx + k = 0$.

Given that $x^2 + 8kx + k = 0$ has equal roots,

b find the value of k

c for this value of k, find the solution of the simultaneous equations.

9 A swimmer dives into a pool. Her position, p m, underwater can be modelled in relation to her horizontal distance, x m, from the point she entered the water as a quadratic equation $p = \frac{1}{2}x^2 - 3x$. The position of the bottom of the pool can be modelled by the linear equation $p = 0.3x - 6$. Determine whether this model predicts that the swimmer will touch the bottom of the pool.



Recall that a **set** is a **collection of values** such that:

a) The **order of values does not matter**.

b) There are **no duplicates**.

Recap from GCSE:

- We use curly braces to list the values in a set, e.g. $A = \{1, 4, 6, 7\}$
- If A and B are sets then $A \cap B$ is the **intersection** of A and B, giving a set which has the elements in A **and** B.
- $A \cup B$ is the **union** of A and B, giving a set which has the elements in A **or** in B.
- $\varnothing$ is the empty set, i.e. the set with nothing in it.
- Sets can also be infinitely large. $\mathbb{N}$ is the set of natural numbers (all positive integers), $\mathbb{Z}$ is the set of all integers (including negative numbers and 0) and $\mathbb{R}$ is the set of all real numbers (including all possible decimals).
- We write $x \in A$ to mean "x is a member of the set A". So $x \in \mathbb{R}$ would mean "x is a real number".

$\{1, 2, 3\} \cap \{3, 4, 5\} = \{\mathbf{3}\}$

$\{1, 2, 3\} \cup \{3, 4, 5\} = \{\mathbf{1,2,3,}$

$\{1, 2\} \cap \{3, 4\} = \varnothing$

It is possible to construct sets without having to explicitly list its values. We use:

The | or : means "such that".

$\{expr \mid condition\}$

or $\{expr : condition\}$

Can you guess what sets the following give?

(In words "All numbers $2x$ such that x is an integer)

$\{2^x : x \in$

$\{xy : x, yareprime\} =$

We previously talked about 'solutions sets', so set builder notation is very useful for specifying the set of solutions!

Can you use set builder notation to specify the following sets?

All odd numbers.

All (real) numbers greater than 5.

All (real) numbers less than 5 **or** greater than 7.

All (real) numbers between 5 and 7 inclusive.

Inequality	Solution Set
$2x + 1 > 5$	
$3(x - 5) \geq 5 - 2(x - 8)$	
$-x \geq 2$	

Combining Inequalities:

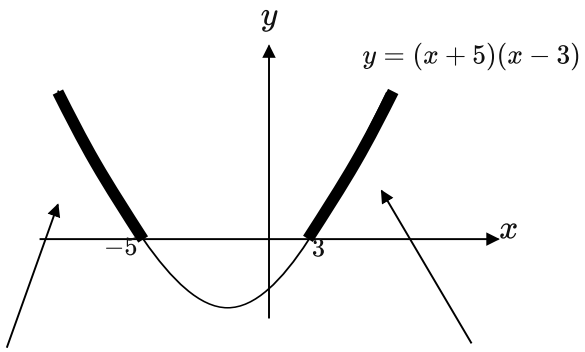
If $x < 3$ and $2 \leq x < 4$, what is the combined solution set?

Solve $x^2 + 2x - 15 > 0$

Step 1: Get 0 on one side

Step 2: Factorise

Step 3: Sketch and reason



Since we sketched $y = (x + 5)(x - 3)$ we're interested where $y > 0$, i.e. the parts of the line where the y value is positive.

Click to bolden >

What can you say about the x values of points in this region?

What can you say about the x values of points in this region?

Solve $x^2 + 2x - 15 \leq 0$

Step 1: Get 0 on one side

$$(x + 5)(x - 3) \leq 0$$

Step 2: Factorise

Step 3: Sketch and reason

Note: we need $\leq$ rather than $<$ to be consistent with the original inequality.

Solve $x^2+5x \geq -4$

Solve $x^2<9$

Note: do not skip the 'sketch step'.

To use the quadratic inequality solver on ClassWiz, go to Menu → Inequalities, then choose 'order 2' (i.e. quadratic)
Try the same on a CG50.

Find the set of values of x for which
(a) $3(x-2) < 8-2x$, (2)
(b) $(2x-7)(1+x) < 0$, (3)
(c) both $3(x-2) < 8-2x$ and $(2x-7)(1+x) < 0$. (1)

Given that the equation $2qx^2 + qx - 1 = 0$, where q is a constant, has no real roots,
(a) show that $q^2 + 8q < 0$. (2)
(b) Hence find the set of possible values of q. (3)

Note: the original equation has no solutions, but the inequality $q^2 + 8q < 0$ *did* have solutions. But we have found the solutions for q that result in the original equation not having any solutions for x. These are different variables, so have different solutions sets, even if the solution set of q influences the solution set of x.

Find the set of values for which $\frac{6}{x} > 2, \quad x \neq 0$

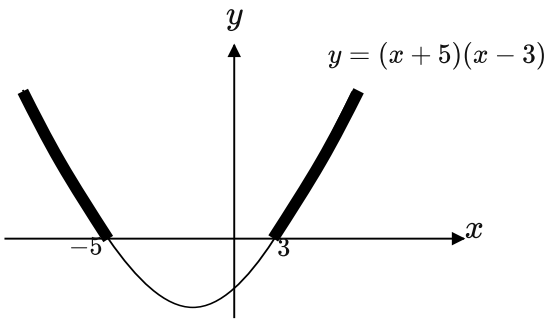
3 Use set notation to describe the set of values of x for which:

g $4(3x + 7) \leq 20$ or $2(3x - 5) \geq \frac{7-6x}{2}$

6 Find the set of values of x for which
$x^2 - 5x - 14 > 0$.

8 Given that $x \neq 3$, find the set of values for which $\frac{5}{x-3} < 2$.

9 The equation $kx^2 - 2kx + 3 = 0$, where k is a constant, has no real roots. Prove that k satisfies the inequality $0 \leq k < 3$.

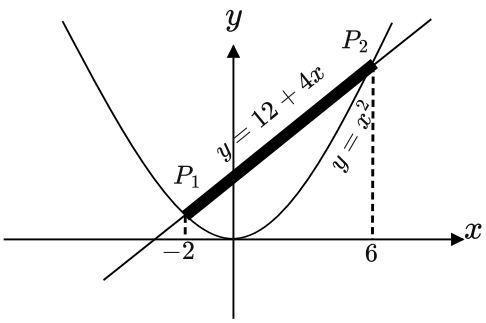


When we solved quadratic inequalities, e.g. $(x+5)(x-3) > 0$
We plotted $y = (x+5)(x-3)$ and observed the values of x for which $y > 0$.
Can we use a similar method when we don't have 0 on one side?

Example: L_1 has equation $y = 12 + 4x$. L_2 has equation $y = x^2$.
The diagram shows a sketch of L_1 and L_2 on the same axes.

a) Find the coordinates of P_1 and P_2, the points of intersection.

b) Hence write down the solution to the inequality $12 + 4x > x^2$.



a

b

On graph paper, shade the region that satisfies the inequalities:
$2y + x < 14$
$y \geq x^2 - 3x - 4$

You did this at GCSE, the only difference here being that the graphs involved might not be straight lines.

Step 2:
An inequality involving x and y represents a 2D region in space. Identify the correct side of each line each inequality represents.

Click to sketch >

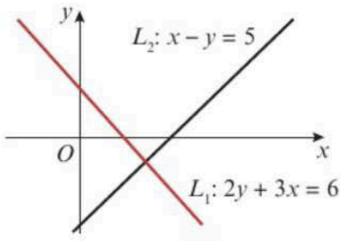
Tip: To quickly sketch $2y + x = 14$, consider what happens when x is 0 and when y is 0.

Tip: Make sure y is on the side where it is positive.
If y is on the smaller side, you're below the line.
If y is on the greater side, you're above the line.

1 L_1 has equation $2y + 3x = 6$. L_2 has the equation $x - y = 5$. The diagram shows a sketch of L_1, L_2.

a Find the coordinates of P, the point of intersection.

b Hence write down the solution to the inequality $2y + 3x > x - y$.



3 Find the set of values of x for which the curve with equation $y = \mathrm{f}(x)$ is below the line with equation $y = \mathrm{g}(x)$.

a $\mathrm{f}(x) = 3x^2 - 2x - 1$, $\mathrm{g}(x) = x + 5$

b $\mathrm{f}(x) = 2x^2 - 4x + 1$, $\mathrm{g}(x) = 3x - 2$

c $\mathrm{f}(x) = 5x - 2x^2 - 4$, $\mathrm{g}(x) = -2x - 1$